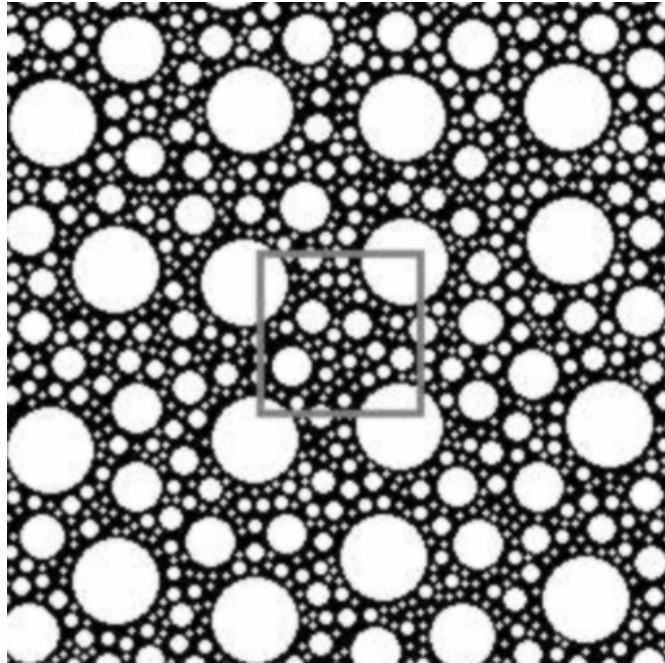




Speckle Mimic Literature Reviews

Lavya G

September 2026

Contents

1	Introduction	1
2	Speckle Quality Assessment	2
2.1	Overview	2
2.2	Speckle Generation Methods	2
2.2.1	Polydisperse Random Sequential Adsorption	2
2.2.2	Multi-Metric Optimization	3
2.3	Speckle Fabrication Methods	3
2.3.1	Constructive Methods	4
2.3.2	Destructive Methods	7
2.3.3	Focused Ion Beam: Deposition and Milling	8
2.3.4	Which Method to Use?	8
2.4	Speckle Assessment Metrics	9
2.4.1	What is a Good Speckle Pattern?	9
2.4.2	Local Parameters	9
2.4.3	Global Parameters	12
2.4.4	Correlation and Sensitivity	14
2.4.5	Outlook	14
2.5	Speckle Statistics	14

Chapter 1

Introduction

blah blah blah.

Chapter 2

Speckle Quality Assessment

2.1 Overview

This chapter is all about trying to understand how to describe and create various speckle patterns. This includes going over basic speckle quality assessment metrics (both local and global) and their fabrication methods. The purpose of this whole section is to understand how to construct an effective cost function for preserving speckle statistics.

A lot of the information so far has been obtained by working through Dong et al.'s review paper on speckle pattern fabrication and assessment [1].

Since speckle patterns are such a central part of DIC, it makes sense to split the discussion into a few pieces: how we assess the quality of a speckle pattern, how we physically fabricate one, and how we might generate one algorithmically. Note also that the speckle pattern is intricately tied to the DIC subset size through the optimal subset size choice, since a smaller speckle size will allow for a smaller subset but will perform worse on a larger subset size. There have been studies on this relationship (see [2]) that are quite convincing.

2.2 Speckle Generation Methods

This section talks about some ways that we can generate speckle patterns computationally, rather than fabricate them physically.

2.2.1 Polydisperse Random Sequential Adsorption

This is a general computational physics algorithm which finds some applications in DIC for speckle generation. It was used for speckle generation in [3].

The pattern generator does not generate samples in a very trivial way. The basic idea is to first start by generating samples with a diameter d_{max} , distributed in the free space such that there is no overlap.

After that, a smaller diameter size is selected and more speckles are generated in whichever bit of the domain is left untouched by other speckles (again, no overlap).

This shrinking process is continued until a minimum diameter d_{min} is reached. The resulting speckle patterns from this algorithm look like:

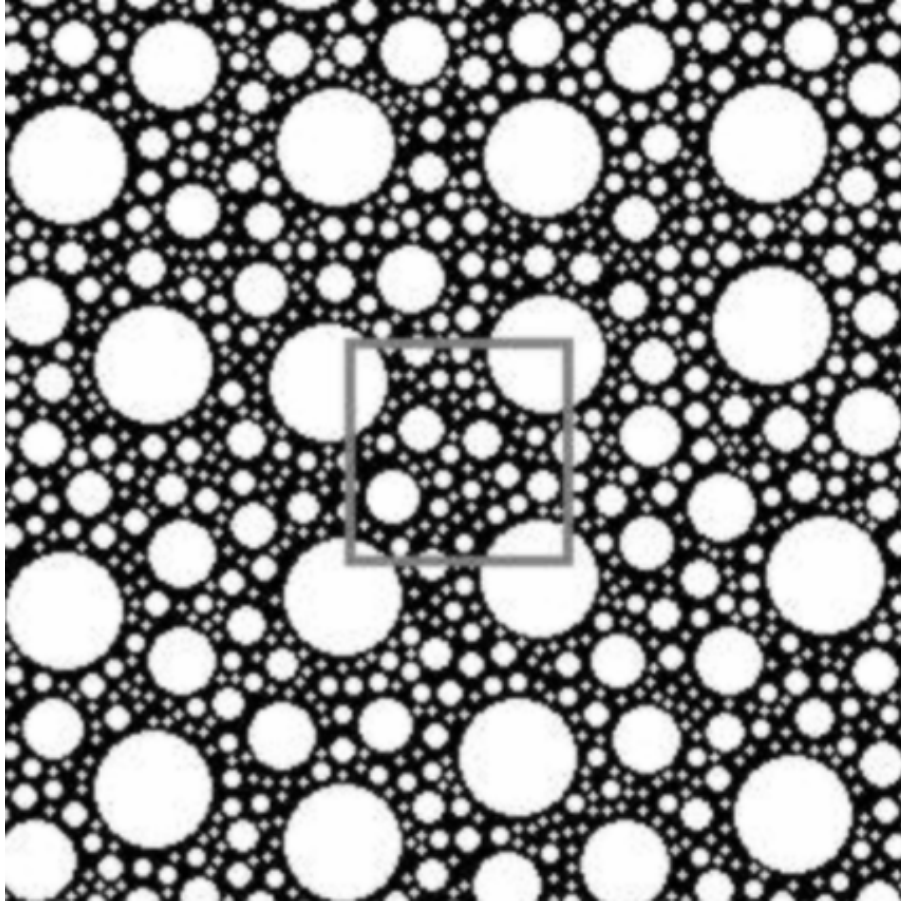


Figure 2.1: Speckle pattern generated via Polydisperse Random Sequential Adsorption.

2.2.2 Multi-Metric Optimization

This is a way of speckle generation used in DIC. It involves using a combination of various speckle assessment metrics to recreate a pattern for a fixed subset size.

This method was proposed by Bomarito et al. [4]. It is interesting because it treats speckle pattern generation as a multi-metric optimization problem, choosing some combination of metrics since each speckle assessment metric comes with its own drawbacks. I am not writing too much in detail about this method but it is worth looking at the original paper.

2.3 Speckle Fabrication Methods

One of the methods to imprint a speckle pattern on a sample is simply by using an airbrush to make white or black dots. We can also use a laser to imprint the speckle pattern (optically) on the surface. These are just some of the types of fabrication available to the experimentalist.

Note: These are definitely not all possible methods! There is a lot of literature around so it is something to keep updating with time.

Artificial speckle pattern (the black and white dots) fabrication can be broadly split into **constructive** and **destructive** methods. Airbrushing, etc. can be considered a constructive method while scratching is considered a destructive method.

2.3.1 Constructive Methods

The first method below (spraying) is used for mm to meter range. The ones following are used in the nano to micro scale.

Airbrushing / Spraying

This is a method that can be used to generate speckle patterns well in the macro (m to mm) scale. The method is simply to take an airbrush and draw out the speckle pattern. There are however some parameters to pay attention to: nozzle diameter, distance between substrate and nozzle, air pressure, and solution viscosity. These parameters can influence the speckle size distribution, and a pre-experiment should be done to determine them. This method gets a bit more complicated to make successful at higher temperatures because standard white and black paints may not work, and there is also a lot of thermal disturbance at these temperatures. There is research being done on fabricating micro-scale patterns of a few μm .

Advantages: very easy to operate; very cheap to run.

Disadvantages: conventional speckle paints and toners can sometimes show specular reflections (especially on moist surfaces), which can be mitigated using cross-polarization; there are a lot of parameters with an airbrush that need to be studied so that the impact on the speckle size distribution is known.

Spin Coating

This is a constructive method that can be used to create speckle patterns in DIC. However, like most of the other methods, it has applications outside of DIC. The basic stages of this method can be seen in the figure below.

This method uses centrifugal force to distribute microscopic particles across a surface. First, we mix some sort of microscopic particle into a liquid solvent. This solution is then dispensed onto the sample where it is spun at high speeds for a random dispersion. Most of the liquid is lost but whatever is left is a very thin film.

The liquid solvent is intentionally chosen to be volatile so it evaporates off while the sample is spinning, so what is left afterwards is just the particles imposed upon the substrate. There are ways that we can prevent the resulting speckle pattern from falling off; these are chemical in nature and not something I plan on covering here.

Advantages: abundant literature (well researched); low cost; fast operation.

Disadvantages: large substrates can't be spun fast enough to let the film thin out. Even these methods have different parameters to tweak, such as particle concentration, particle type (to control speckle size distribution), and centrifugal run times/speeds.

Compressed Air Technique (CAT)

This is a technique for speckle fabrication in DIC. As always, this may not be the only application of this technique. The basic idea of this technique can be seen in the figure below.

Compressed air enters through the bottom of the chamber. There is a powder comprised of micro-scale particles below the inlet. The compressed air blows these particles upwards through a series of filters, which are needed to make sure no agglomerated particles make it through.

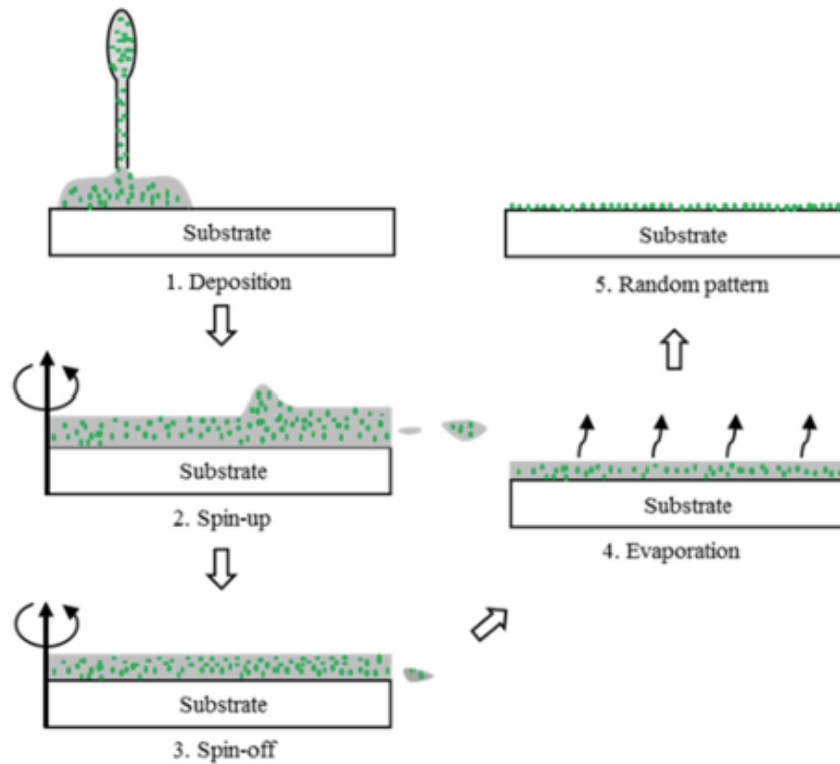


Figure 2.2: Stages of the spin coating process.

The micro particles adhere to the specimen using Van der Waal's forces. The filter configuration can be tweaked to give us the optimal pattern.

One big disadvantage of this method is the fact that it relies purely on Van der Waal's forces to make the particles adhere. This stuff will also get weird at high temperatures so a lot more research is needed with these methods.

Nano-Film Remodeling

This method is used for speckle fabrication at the micro scale, though it may have applications beyond DIC. The basic idea is to use vapours of some volatile solvent systematically to remodel thin films. The film morphology (structure) can be controlled by choosing parameters such as initial film thickness, exposure time, and substrate temperature. This method works very nicely at the nanometer scale, which is quite impressive.

There are some issues with coating the film on the surface as well, and the different film coating methods could switch things up a little bit. This is (from what I can tell) a new, WIP method, so it is not as explored as something like spin coating, which has been around for about 50 years.

Lithography

This is a method for speckle pattern generation used in DIC, but may have other applications, such as the manufacture of integrated circuits and micro-scale devices. We use electron-beam lithography to generate the micro/nano-scale patterns; the basic steps are illustrated below.

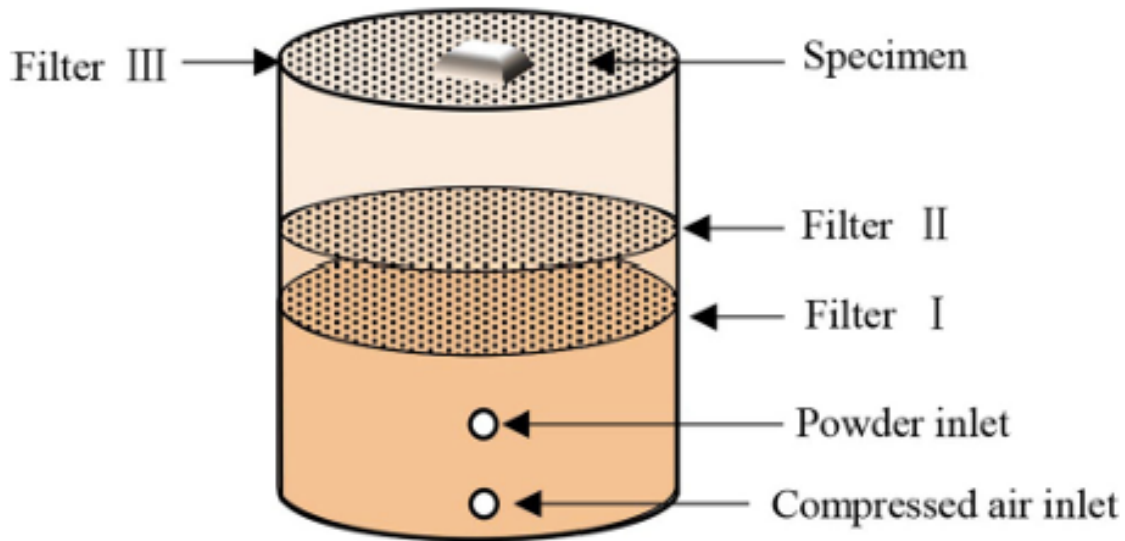


Figure 2.3: The Compressed Air Technique chamber setup.

First, some PMMA “resist” is coated onto the substrate via spin coating. Then the resist is irradiated by electron beams. The irradiated portions of resist are dissolved, since the electron beam weakens the chemical bonds in that area and lets the portion get dissolved away, leaving resist holes where the electron beams hit.

The specimen is then sprinkled with material fragments. The way these bond together is interesting and works at an atomic level: the particles hit the substrate and knock other bits off, forming chemical bonds with the substrate. After the sprinkling is done, the remaining resist is dissolved away.

This method has been shown to have good thermal stability in some research, but its use at higher temperatures has more room for research. It struggles with high cost and applicability to non-flat substrates, as the pattern generation depends on the working distance.

Comparison of Constructive Methods

Airbrushing and spraying is very cost effective and easy to do, though it has a lot of parameters to mess around with and can bring about a good bit of specular reflection due to conventional paints.

Spin coating is cheap, well developed, and fast, but can’t be applied to big substrates.

The compressed air technique has some benefits, but the bonds are a bit weird because they rely purely on Van der Waal’s forces.

Nano-film remodeling is quite complicated as a method and is still being researched. The bonding is multi-stage, as the film has to be remodeled separately and then bonded on.

Lithography is independent of substrate, repeatable, and designable, but it is very expensive and can’t be applied to non-flat substrates.

FIB is expensive but very controlled and repeatable; one big thing about FIB deposition is that it may do some surface damage due to high ion beam energy.

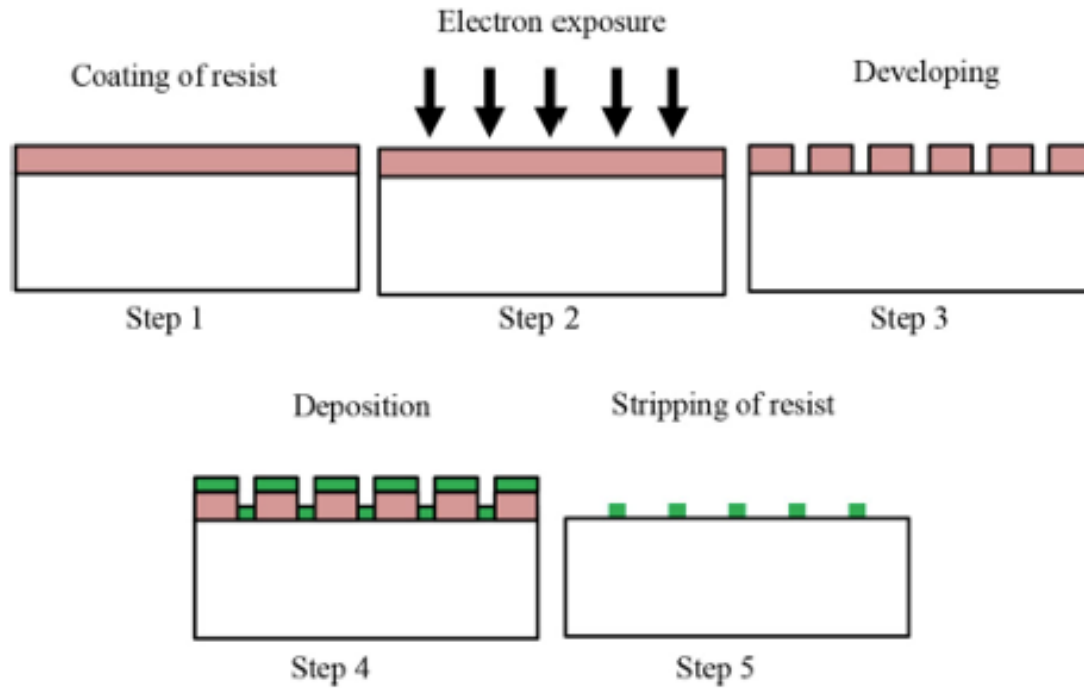


Figure 2.4: Steps of the electron-beam lithography process.

2.3.2 Destructive Methods

All of the destructive methods below can be used in micro-scale pattern fabrication. They share a common disadvantage: the specimen will always be damaged, which may cause some sort of residual stresses that are not ideal.

Scratching and Abrading

This is a method of speckle fabrication used in DIC. The idea is very simple: take some type of scratching tool and scratch out a nice little pattern.

What this type of scratching does is change surface roughness at points, which can allow us to change how much specular versus diffuse reflection occurs at a point, thus changing the grey level. Specular reflections as a whole are pretty bad for DIC, which is an interesting aspect of this method, though if we use cross-polarization we can minimize the damage of this method in that sense.

This method, similar to spraying, is very cheap and easy to do. However, here, the specimen might actually get damaged and we will get some residual stresses. Additionally, if we fail to scratch out the pattern right, the specimen is just wasted.

Chemical Etching

Focused Ion Beam Milling

See Section 2.3.3 below.

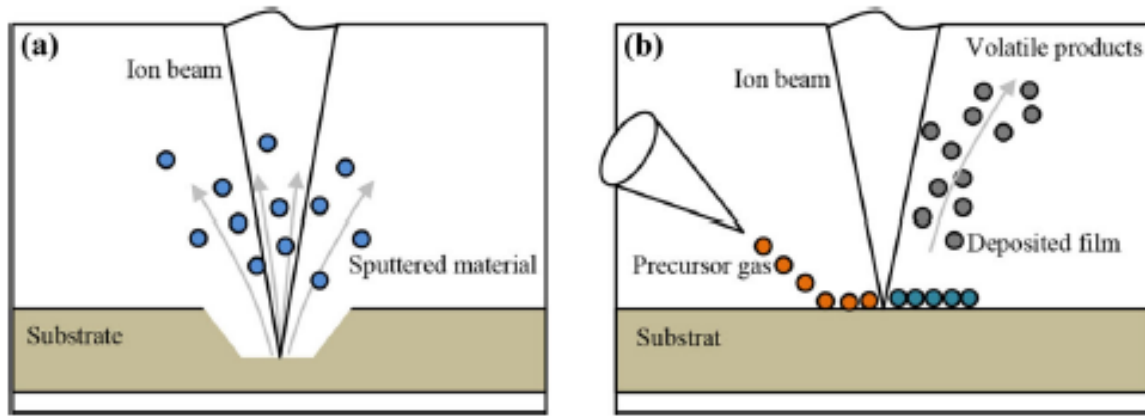


Figure 2.5: FIB deposition and FIB milling processes.

Comparison of Destructive Methods

Scratching is a very cheap way of getting things done and is quick, though failure to scratch properly means the specimen is damaged.

FIB milling has similar advantages to FIB deposition; the additional thing here is that we need a Scanning Electron Microscope (SEM) to make it work.

2.3.3 Focused Ion Beam: Deposition and Milling

This is a technique for speckle fabrication and can be used for DIC. There are two ways of using the ion beam: constructively or destructively. The constructive method is called FIB deposition, while the destructive method is called FIB milling. Both of them are shown in the figure below.

FIB Milling works in tandem with an SEM. The ion beam drills into the substrate to create a speckle pattern based on depth; since we are using an SEM, the depth registers as a different gray level and we get a nice speckle pattern.

FIB Deposition: first we release a precursor gas on the sample surface, which is adsorbed on surface molecules. The ion beam then decomposes this gas, such that the volatile products get exhausted out due to vacuum while the desired reaction products stay on the sample surface as a thin film. The desired products stay on the surface chemically (just like in lithography), though here the high energy of the ion beam creates a mixing zone where the substrate is knocked away.

This method has a very quick processing time but can actually damage the surface. There are also a bunch of parameters that influence this fabrication, like the total deposition time, speckle temperature, ion beam current, and dwell time. These ion beam methods are also very expensive (obviously).

2.3.4 Which Method to Use?

The following is a summary drawn directly from the review paper.

“Artificial techniques of spraying and airbrushing are inexpensive and can be easily operated. For ordinary testing conditions such as tensile, compressive experiments at room temperature and elevated temperatures, this patterning

technique is recommended. Micro/nano-scale speckle patterning techniques including spin coating, compressed air technique, nano-film remodeling, FIB and lithography are mainly used for the high magnification optical microscopes, SEM, AFM, etc. Scratching/abrading is simple and easily operable, but is surface damaging. Spin coating, compressed air technique and nano-film remodeling are potential techniques due to their low cost and high efficiency. FIB and lithography are usually expensive and time consuming.” [1]

2.4 Speckle Assessment Metrics

This section is about understanding how to check the quality of a speckle pattern. We can start by doing a quick qualitative analysis of what is expected, then branch off into different parameters of the speckle pattern.

2.4.1 What is a Good Speckle Pattern?

To improve DIC accuracy, the speckle pattern must have high contrast, randomness, isotropy (no directionality), and stability. The first three are important to keep the features unique; the fourth one is so that the speckle pattern actually sticks well to the surface and deforms accordingly, not separately.

Speckle quality assessment metrics are given as a series of parameters, split into local and global parameters.

2.4.2 Local Parameters

In contrast to global parameters, local parameters quantify speckle pattern quality with individual subsets in mind. These are expected to help find the optimal subset size. The problem with these parameters is that they are, well, local: they don’t really show how quality differs between subsets and are very focused on a singular subset.

The most widely adopted local parameter is SSSIG. This parameter has been derived with considerable theory in mind, in contrast to the others, which are more empirical. The other parameters can all be “derived” somehow from SSSIG, so it makes sense to mostly focus on that.

Sum of Square of Subset Intensity Gradients (SSSIG)

We let S represent the subset domain (i.e., the subset size). Then x, y become the coordinates of each individual pixel, and $f(x, y)$ is the intensity (gray-level) function. It is standard notation to define:

$$f_i(x, y) = \partial_i f(x, y)$$

With this notation, the sum of square of subset intensity gradients becomes:

$$SSSIG = \sum_{x, y \in S} (f_x(x, y)^2 + f_y(x, y)^2)$$

To understand the impact of this parameter, it is worth covering its derivation as done in [5].

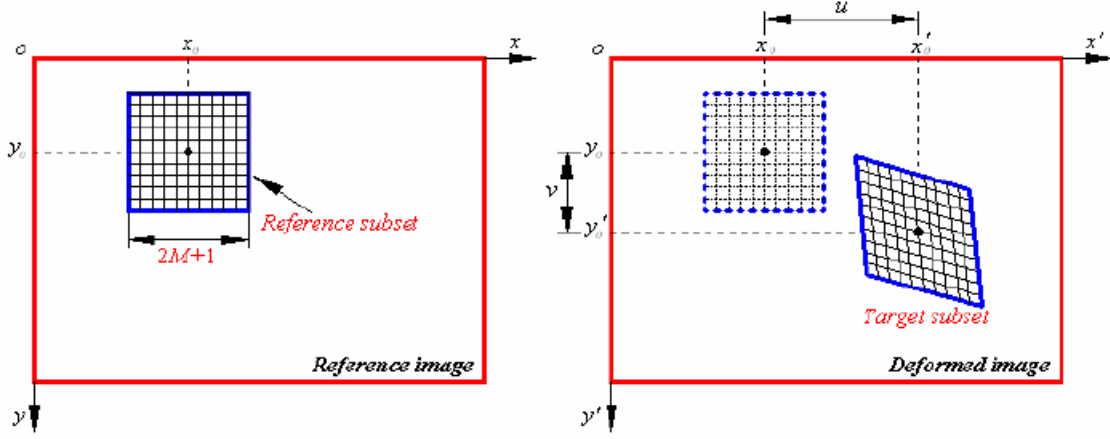


Figure 2.6: Subset notation used in the SSSIG derivation.

Derivation of SSSIG The figure above is good to get the notation right. We start with a subset made of $2M + 1$ pixels on each side, and can start by looking at the correlation function; for this model we consider the sum of squared differences (SSD) correlation criterion:

$$C(\Delta u, \Delta v) = \sum_{x=-M}^{x+M} \sum_{y=-M}^{y=M} [f(x, y) - g(x', y')]^2$$

Here, $f(x, y)$, $g(x', y')$ are the intensity profiles for the reference and deformed images. We assume a very simple (zeroth-order) subset shape function to keep the mathematical model very simple. This creates displacements like:

$$x' = x + u = x + u_0 + \Delta u$$

where we have split the pixel and subpixel accuracies, the former assumed known from a previous rough optimization loop. This shape function can be expanded in a Taylor series, which implies that this model will break down for huge subsets:

$$g(x + u_0 + \Delta u, y + v_0 + \Delta v) = g(x + u_0, y + v_0) + \Delta u \cdot g_x(x + u_0, y + v_0) + \Delta v \cdot g_y(x + u_0, y + v_0)$$

These gradients can be calculated via central difference. Minimizing the correlation function C yields expressions for Δu and Δv :

$$\Delta u = \frac{\sum \sum (g_y)^2 \sum \sum [(f - g)g_x] - \sum \sum (g_x g_y) \sum \sum [(f - g)g_y]}{\sum \sum (g_x)^2 \sum \sum (g_y)^2 - (\sum \sum (g_x g_y))^2}$$

with a similar expression for Δv . To gauge accuracy, we need to account for noise, which we can add as:

$$f'(x, y) = f(x, y) + \eta_1(x, y), \quad g'(x, y) = g(x, y) + \eta_2(x, y)$$

We assume the noise has a constant mean value, so for $\eta = \eta_1 - \eta_2$, $E(\eta) = \text{const}$. We also assume the noise values at different pixels are independent. If not, the statistical analysis here would be wrong, meaning this analysis **does not apply to noise that is not stochastic**. The final assumption is that the noise levels are small compared to the

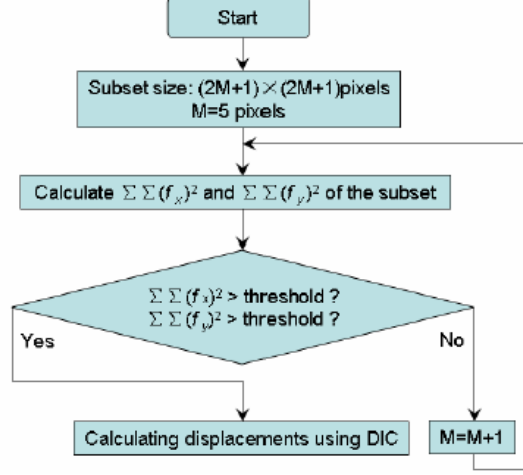


Figure 2.7: Threshold-based subset size selection algorithm using SSSIG.

gray levels. With these approximations we have:

$$\Delta u' = \Delta u - \frac{\sum g_y^2 \sum (\eta g_x) - \sum (g_x g_y) \sum (\eta g_y)}{\sum g_x^2 \sum g_y^2 - (\sum g_x g_y)^2}$$

Notice how the noise term here has squared gradients as the denominator. Taking the expectation and variance of this delta:

$$E(\Delta u') = \Delta u - N \cdot \frac{\sum g_y^2 \sum g_x - \sum (g_x g_y) \sum g_y}{\sum g_x^2 \sum g_y^2 - (\sum g_x g_y)^2}$$

where N is the mean noise value taken out of the expectation. For the variance, assuming $g_x = f_x(x, y)$ (i.e., the intensity gradients don't change much between correlated subsets):

$$D(\Delta u') \approx \left(\frac{D(\eta)}{\sum \sum (f_x)^2} \right)^{1/2}$$

This shows how the impact of noise on correlated displacement readings reduces if we have a high SSSIG and a low $D(\eta)$, N .

It is possible in DIC experiments to get the noise variance by taking some static images. The mean and variance of the noise are generally pretty fixed, which means measurement quality is improved by improving the SSSIG. This parameter can also be used to design a threshold-based algorithm for choosing subset size, as shown below.

The thing with SSSIG is that it is unable to differentiate between random speckle patterns and periodic patterns, since there is no uniqueness encoded into it. It can't be used in isolation; this is combined with the general drawbacks of local parameters.

Subset Entropy

Consider a subset of size $M \times N$, where we define:

$$\delta = \frac{\sum_{P \in S} \sum_{i=1}^8 |I_P - I_i|}{2^\beta M N}$$

Here, P is any pixel point inside the subset while δ is the subset entropy, summing the intensity differences of a pixel compared to the 8 pixels surrounding it. This metric was proposed in [6].

This metric is derived empirically, so it doesn't have that much theoretical backing, though it could be a decent statistic for a speckle pattern. It suffers from the issues that local parameters have in general, and it's easy to see that this metric biases you towards having a speckle size of just 1 pixel, a problem worth being aware of.

Mean Subset Fluctuations

Unfortunately, the original paper for this metric was taken down, so I have had to use an LLM to fill in the blanks here. It's the same idea as subset entropy and thus suffers the same biases and drawbacks. For local parameters in general, SSSIG is the way to go.

2.4.3 Global Parameters

Global parameters look at the entire speckle pattern instead of just a few subsets, and are more commonly used as an indicator of overall pattern quality.

Speckle Size Distribution

It can be seen from [2] that the mean speckle size is very intricately linked to the subset size, and the two go hand in hand. If your speckle pattern loses the mean speckle size, things would go bad with your actual validation.

The use of mean speckle size as a metric comes with a drawback though: it is a very shortsighted metric because it does not segregate against patterns, so even a uniform or directional pattern is treated the same as a truly randomly sampled pattern.

Mean Intensity Gradient (MIG)

This is a global generalization of SSSIG. In SSSIG, you compute gradients over a singular subset, which comes with the drawback of not looking at the global picture of the speckle pattern. The alternative, proposed by Pan et al., is the mean intensity gradient, basically SSSIG over the whole domain:

$$MIG \equiv \delta_f = \frac{\sum_{i=1}^W \sum_{j=1}^H |\nabla f(x_i, y_i)|}{W \times H}$$

This means it suffers from the same biases and drawbacks as SSSIG: it doesn't show variability over a pattern, and gradient computation can be time consuming over a whole domain.

These drawbacks are not to say the metric is useless; it is a generalized version of SSSIG, which gives it inherent power. Numerical experiments in [7] show that this metric does make a difference and should ideally be maximized.

Shannon Entropy

This concept originates from information theory but has found its application as a global parameter for speckle assessment. It tells you about how much information is stored in

an image:

$$H(Y) = - \sum_{j=0}^{2^\beta-1} p(a_j) \log(p(a_j))$$

Here, Y represents the image/speckle pattern itself, and $p(a)$ is the normalized probability density of each gray level. This metric gives a very good high-level picture of the pattern, but it fails to properly show the fundamental features of the pattern. A good quality speckle pattern should ideally have a large Shannon entropy. The work here was initially done in [8], which shows pretty good numerical experiments for this metric.

Autocorrelation Peak Sharpness

This requires an understanding of how DIC sensitivity and precision can be formulated in terms of correlation (see Section 2.4.4 below), and involves autocorrelation, which is used frequently in Monte Carlo methods.

The sensitivity in DIC can be understood using the gradient of correlation in shifted images (in DIC, correlation is normalized so that fully correlated images are 0). To assess the quality of a single pattern, it is useful to take just one image and artificially shift it. We start by measuring the autocorrelated value with a 0px shift (A_0), then shift by 1px in the four cardinal directions to get a mean autocorrelation shift $A_{\pm 1}$, then shift based on full contrast random noise to get A_∞ . The autocorrelation peak sharpness is then:

$$R_{peak} = \sqrt{\frac{A_\infty - A_0}{A_{\pm 1} - A_0}}$$

This measures how sharp the peak of the autocorrelation curve is: a lower radius means a better speckle pattern, and the math here amounts to fitting a parabola to the peak.

The problem with this metric is that it is biased towards tiny checkerboard patterns and doesn't track the randomness of the pattern. It also only looks locally at a single peak, with no way to know how many secondary peaks there are, and for that reason the autocorrelation peak comparison metric (below) might be better suited.

Autocorrelation Peak Comparison

This metric is similar to autocorrelation peak sharpness, except instead of measuring the sharpness of a single peak, we first use autocorrelation to find the ‘‘correlation landscape.’’ This metric was first introduced in [3].

Based on the correlation values obtained (C), the metric q is:

$$q = \frac{C_{max} - C_{sec}}{\sigma(C)}$$

where C_{max} is the correlation value for the main peak, C_{sec} is the correlation value for the secondary peak, and $\sigma(C)$ is the standard deviation of these correlations.

This measures uniqueness over a more global landscape than the single-peak metric, though it incurs a heavier computational cost. It also weights a single large secondary peak as worse than one with slightly smaller but many other secondary peaks, making the problem more nonconvex.

Watershed Radius

This metric deals with the robustness of initial guesses in DIC correlation. We can use autocorrelation as a surrogate for future correlations, informing a watershed algorithm to understand which initial guesses will deviate significantly from the true solution. What we want for a speckle pattern is a large radius where we are within a safe region for the algorithm to converge:

$$R_W = \sqrt{\frac{W}{\pi}}$$

where W is the area of the displacements in the watershed of the primary autocorrelation peak.

2.4.4 Correlation and Sensitivity

In DIC, we can define a percentage image correlation by taking the reference image and a perturbed image. Instead of moving our subset into the frame of the speckle pattern by optimizing the correlation function, we can keep the frame constant and look at how un-correlated a function gets when moved by a certain displacement.

For a very sensitive model, a small displacement should instantly decorrelate the speckle patterns below a certain threshold, so the sensitivity of the model changes with respect to the gradient of decorrelation against displacement. This perspective can be seen in [9] (assuming we have accounted for the stochastic noise of the process).

If we do virtual shifts instead of taking actual shifts through multiple images, we can work with only one image, and this is what we call autocorrelation.

2.4.5 Outlook

Overall, according to [1], the usage of MIG and SSSIG is the most likely candidate pairing for proper quality identification. However, these metrics by themselves do not properly quantify the randomness of a pattern, which is why no single parameter is good enough by itself, and a good idea is to use a bunch of metrics in tandem.

The table below, from [1] (Check the reference or obsidian), summarizes the advantages and disadvantages of each metric, though in the quest for compactness it loses a lot of key information.

2.5 Speckle Statistics

This section will cover the deeper statistical treatment of speckle patterns, building on the assessment metrics above, once the corresponding literature review is complete.

Bibliography

- [1] Y. Dong and B. Pan, “A Review of Speckle Pattern Fabrication and Assessment for Digital Image Correlation,” en, *Experimental Mechanics*, vol. 57, no. 8, pp. 1161–1181, Oct. 2017, ISSN: 1741-2765. DOI: 10.1007/s11340-017-0283-1 Accessed: Sep. 7, 2026. [Online]. Available: <https://opg.optica.org/oe/abstract.cfm?uri=oe-25-24-30259>
- [2] D. Lecompte et al., “Quality assessment of speckle patterns for digital image correlation,” *Optics and Lasers in Engineering*, vol. 44, no. 11, pp. 1132–1145, 2006. DOI: 10.1016/j.optlaseng.2005.10.004
- [3] G. Stoilov, V. Kavardzhikov, and D. Pashkouleva, “A comparative study of random patterns for digital image correlation,” *Journal of Theoretical and Applied Mechanics*, vol. 42, pp. 55–66, Jun. 2012. DOI: 10.2478/v10254-012-0008-x
- [4] G. F. Bomarito, J. D. Hochhalter, T. J. Ruggles, and A. H. Cannon, “Increasing accuracy and precision of digital image correlation through pattern optimization,” *Optics and Lasers in Engineering*, vol. 91, pp. 73–85, Apr. 2017. DOI: 10.1016/j.optlaseng.2016.11.005
- [5] B. Pan, H. Xie, Z. Wang, K. Qian, and Z. Wang, “Study on subset size selection in digital image correlation for speckle patterns,” *Optics Express*, vol. 16, pp. 7037–7048, May 2008. DOI: 10.1364/OE.16.007037
- [6] S. Yaofeng and J. H. L. Pang, “Study of optimal subset size in digital image correlation of speckle pattern images,” *Optics and Lasers in Engineering*, vol. 45, no. 9, pp. 967–974, 2007. DOI: 10.1016/j.optlaseng.2007.01.012
- [7] B. Pan, Z. Lu, and H. Xie, “Mean intensity gradient: An effective global parameter for quality assessment of the speckle patterns used in digital image correlation,” *Optics and Lasers in Engineering*, vol. 48, no. 4, pp. 469–477, 2010. DOI: 10.1016/j.optlaseng.2009.08.010
- [8] X.-Y. Liu, R.-L. Li, H.-W. Zhao, et al., “Quality assessment of speckle patterns for digital image correlation by shannon entropy,” *Optik*, vol. 126, no. 23, pp. 4206–4211, 2015. DOI: 10.1016/j.ijleo.2015.08.034
- [9] S. Bossuyt, “Optimized patterns for digital image correlation,” in *Conference Proceedings of the Society for Experimental Mechanics Series*, vol. 3, Jan. 2013, pp. 239–248. DOI: 10.1007/978-1-4614-4235-6_34